Hard Ice, Soft Ice

by Bill O'Brien

What's the Difference between **Snow** and **Hail?**

Stand outside in either, and you'll soon find out. Snow is soft and fluffy. Hail is hard, and even small hailstones will sting slightly if they strike bare skin. And that's just the usual, pea-sized stones. Some hailstones are like golf balls – just as big and just as hard. The biggest hailstones recorded have been the size of grapefruit or even coconuts.

Hail can be very damaging, especially in orchards when the fruit is becoming soft and ripe. For example, sudden hailstorms have wiped out New Zealand's Christmas cherry crop in the past. On the Great Plains of North America, hail is so damaging that farmers call it "the white plague". And in 1985, several hundred people were killed by a massive hail strike in India.

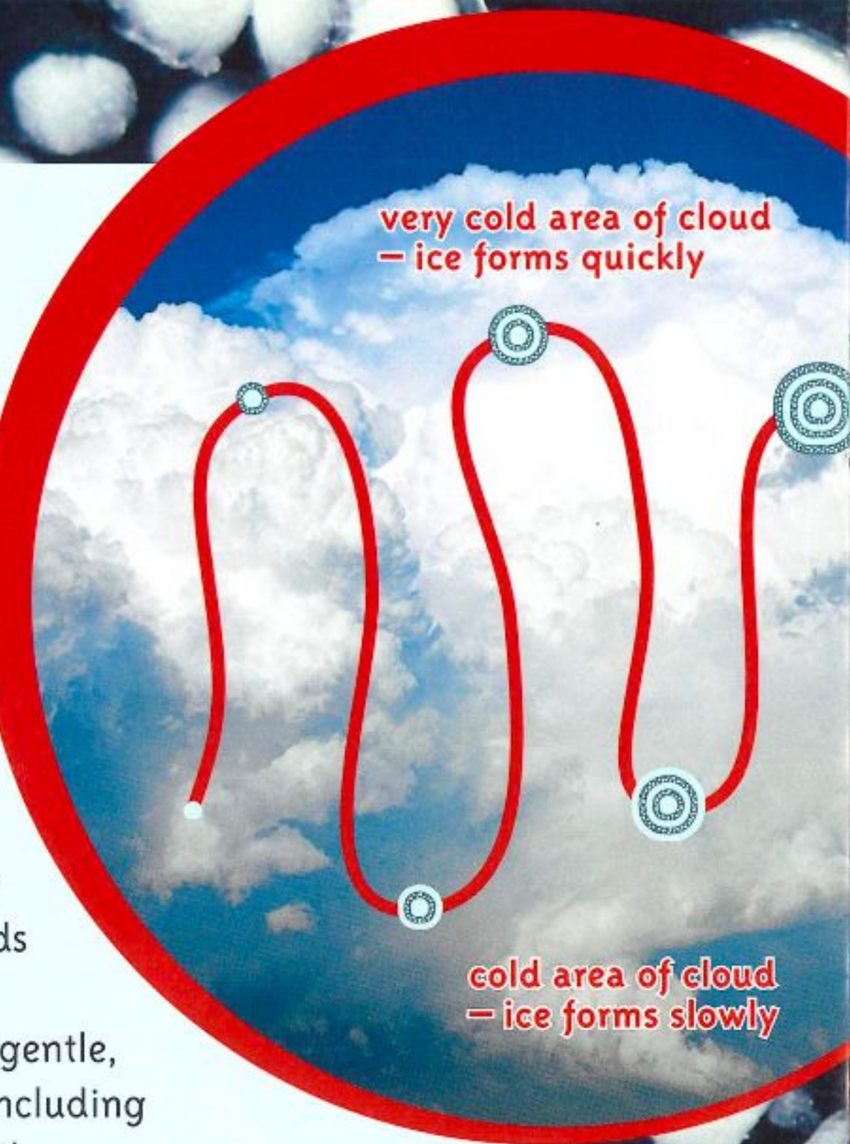
How **Snow** Forms

So why is snow so light and soft compared with hail? Let's think about how a snowflake forms. High up where the atmosphere is very cold, tiny particles of water vapour (a gas) within a cloud condense onto a speck of dust and freeze. But they don't all freeze suddenly into a solid lump of ice. Rather, the particles freeze one by one, and so the crystal builds up gradually into a delicate, feathery form that's a bit like frost. Each crystal looks slightly different, but they're all shaped like a hexagon or a six-pointed star. Although a snowflake grows, it remains very light for its size because there's so much air within its structure. For this reason, snowflakes fall only slowly through the cloud. Sometimes they're lifted up again by rising air. As they fall or rise, they bump into each other and join up into bigger snowflakes. Most grow no bigger than a thumbnail, but some can become the size of your palm.

Eventually a snowflake becomes heavy enough to fall steadily downwards. Often a whole shower of snowflakes drop from a cloud at the same time. Very large, dense clouds can produce so much snow that a blizzard results. Although an individual snowflake is very light, the combined weight of a very heavy snowfall can be enough to snap the branches off trees, cause avalanches, or collapse buildings, power lines, and other structures.

How **Hail** Forms

It has to be a cold day for it to snow. If not, the snowflakes melt and become rain as they fall through warm layers of air. But hail can strike even on very hot days. This is because a hailstone is solid ice, which is less likely to melt away as it falls. To find out why hail's so solid, let's look at how it forms. Hail develops within cumulus clouds. "Cumulus" is Latin for "heap", and this name was given to big, billowing clouds that pile up high like huge stacks of candyfloss. These clouds may appear gentle, but they cause very violent weather, including thunderstorms, torrential rain, and hail.



Cumulus clouds form over the sea when warm air causes sea water to evaporate and form a gas called water vapour. As the water vapour rises, it cools down and begins to condense into billions of tiny, liquid droplets. Water vapour is invisible, and so is an individual droplet because it's so small. But billions of droplets together can be seen as steam, mist, fog, or cloud. When water vapour condenses into liquid water, energy is released in the form of heat. And when a lot of vapour condenses, a huge amount of heat is released. This is just what happens within a cumulus cloud as it develops.

A hailstone can move up and down repeatedly between cold and very cold areas within a cloud. Each time, two layers of ice are added to the hailstone. The first layer is like clear glass. This layer forms in the cold part of the cloud, where water freezes slowly enough for tiny air bubbles to escape from the water, leaving the ice clear. If the hailstone reaches very cold areas higher up in the cloud, a frosty layer of ice forms. This ice is frosty because it freezes very quickly – too quickly for bubbles to escape.

So, making a cloud is hot work! And because hot air rises, updrafts form within cumulus clouds. Travelling at speeds of up to 100 kilometres per hour, these warm updrafts, or "thermals", are very powerful. They carry huge numbers of water droplets up into cooler layers of the atmosphere, where they freeze and form small hailstones. These hailstones are not light and feathery like snowflakes. Rather, they're solid and dense, which means they fall back down quite rapidly. Often, however, a falling hailstone is blown back upwards by the thermals within the cloud. As the hailstone rises through colder parts of the cloud again, new layers of water freeze onto it.

A hailstone can whiz up and down repeatedly, getting bigger and bigger as it goes. Scientists can tell how many times a hailstone has been up and down by slicing through it and counting the layers of ice. With all those layers, hailstones are a bit like onions, except they don't make your eyes water – unless they hit you on the head! Speaking of which, a hailstone eventually becomes so big and heavy that the thermals within the cloud can no longer carry it upwards. When this happens, the hailstone drops to the ground, sometimes from a height of 10 kilometres.

So that's how we get hard ice and soft ice. Because of the way they form, hailstones are hard and dense compared with feathery snowflakes. And, because of its structure, a hailstone falls much more quickly than a snowflake – ouch!